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Cross-Border Banking Exposures and Macroeconomic Spillover Effects: The Case of Kenya

Prepared by Zhongxia Zhang and Francine Nyankiye

WP/26/245

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ABSTRACT: Kenya is a regional financial center in Africa with sizeable banking activities. However, its cross-border banking linkages and the associated macroeconomic spillover effects have not been studied sufficiently. This paper presents facts about international banks' cross-border claims on Kenya, as well as Kenyan banks' increasing exposures abroad. We document the scale, geography, funding source, market structure, and institutional setting of Kenyan banks' presence in countries inside the East African Community. Furthermore, we discuss the authorities' cross-border supervisory efforts and the status of regional banking harmonization. Finally, our empirical analysis shows that Kenya's exports and host countries' banking sector profitability and asset quality are positively associated with Kenyan banks' overseas expansions, which are closely linked to higher local economic growth and inflation. Together with Kenyan banks' large market shares in several host countries, these results underscore the macro-financial relevance of Kenya's regional banking linkages.

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WORKING PAPERS

Cross-Border Banking Exposures and Macroeconomic Spillover Effects: The Case of Kenya

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1. Introduction

Kenya is a regional financial hub in Africa, yet its cross-border banking linkages with the rest of the world have not been fully documented. Kenya is a premier destination for international banks to do business, and the country also serves as the headquarters for some of Africa's most active banks. The country's banking system is connected to global and regional financial systems through two channels. First, international banks maintain claims on Kenya and operate through subsidiaries, branches, and representative offices. Second, Kenyan banks have expanded into neighboring and other African markets.

While Kenyan banks' presence is significant in several countries, few research studies have discussed the macro-financial implications for the Central Bank of Kenya (CBK) and other banking supervisors in cross-border banking supervision and crisis management. Much attention has been devoted to within-country banking sector developments and supervisory matters. Cross-border banking issues, on the other hand, are typically treated as a lower priority in earlier policy research. Two exceptions are Drummond et al. (2015) and IMF (2015), which examine regional banking integration and challenges for cross-border oversight. Given the remarkable increase in Kenyan banks' cross-border activities in the past decade, it is warranted to take a fresh look at patterns of Kenya's cross-border banking linkages.

The factors behind Kenyan banks' overseas expansions and the associated macroeconomic spillover effects have not been sufficiently investigated either. Cross-border banking can support trade, investment, and financial inclusion, but it also creates channels through which shocks may be transmitted across jurisdictions. For example, deterioration in parent bank balance sheets could affect credit supply in destination countries. Prior studies including Mathieu et al. (2019) tend to focus on drivers of cross-border banking in the African region rather than from a specific country. However, understanding the determinants of Kenya's cross-border banking linkages and their macroeconomic spillover effects is crucial for macro-financial surveillance and financial stability in Africa.

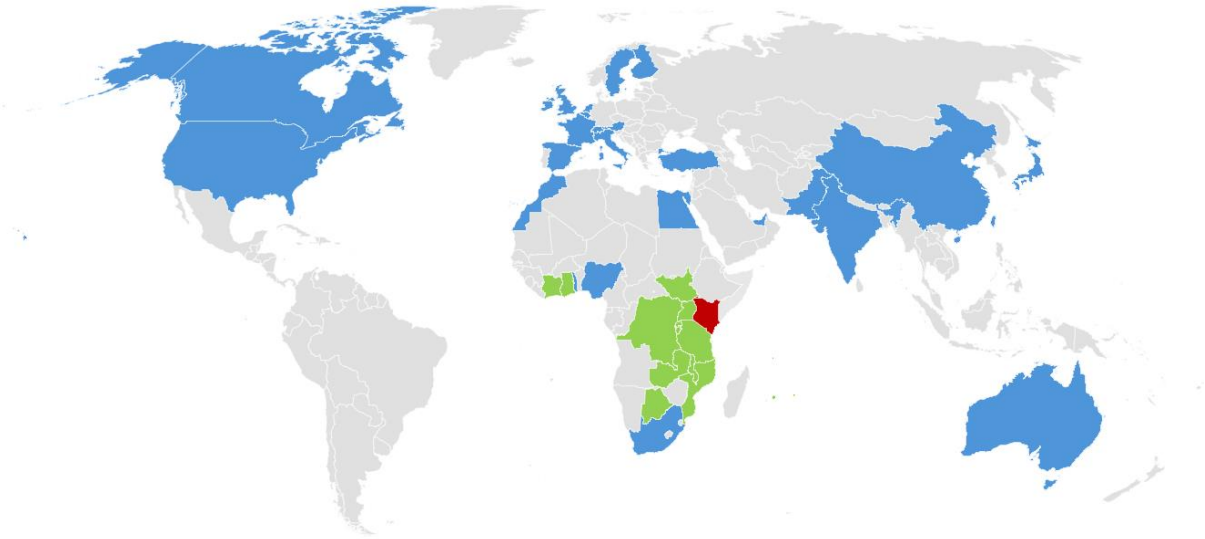
This paper aims to fill the above-mentioned research gaps. We provide comprehensive evidence on rising cross-border claims from international banks into Kenya and Kenyan banks' assets abroad. In particular, Kenyan banks have a significant presence in countries inside the East African Community (EAC) and have also expanded in other parts of Africa. We further present factual observations on the authorities' supervisory efforts and the regional banking integration in the East African Community. Our empirical analysis shows that Kenya's exports to destination countries and banking sector profitability and asset quality in host countries are positively associated with Kenyan banks' overseas expansions. Bank credit expansions go hand in hand with higher real GDP growth and CPI inflation, which highlights the systemic importance of Kenyan banks' cross-border exposures.

The rest of the paper is organized as follows. Section 2 of the paper provides details on international banks' inward cross-border exposures to Kenya and Kenyan banks' outward cross-border exposures to the rest of Africa, as well as cross-border supervisory efforts and regional banking harmonization. Section 3 reveals economic and financial factors that are tied to Kenyan banks' overseas expansions and macroeconomic effects of bank credit growth. Section 4 performs several robustness checks of empirical analysis. Section 5 discusses key findings of the paper and policy implications.

2. Kenya's Cross-Border Banking Exposures

Using the Central Bank of Kenya's banking supervision statistics and the BIS' international banking statistics, our mapping exercise suggests that Kenya has attracted cross-border business from banks located in a vast number of major economies and Kenyan banks have become ubiquitous in the East African Community (Figure 1).

Figure 1. Kenya's Cross-Border Banking Exposures, 2024



Sources: BIS, Central Bank of Kenya, and IMF staff estimates.

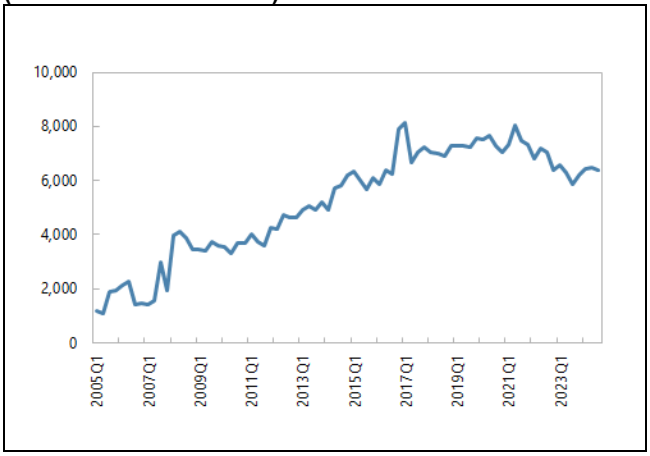
Legend: blue = origin countries of international banks that have claims on Kenya or representative offices in Kenya; green = destination countries where Kenyan banks have operations or ownership of local banks; red = Kenya.

Footnote: The Export-Import Bank of China is considered a bilateral official creditor in the IMF's debt sustainability analysis. The Industrial and Commercial Bank of China owns 19.6 percent of the Standard Bank Group, which is a South African bank that operates in Kenya as Stanbic Bank.

2.1 International Banks' Inward Cross-Border Exposures to Kenya

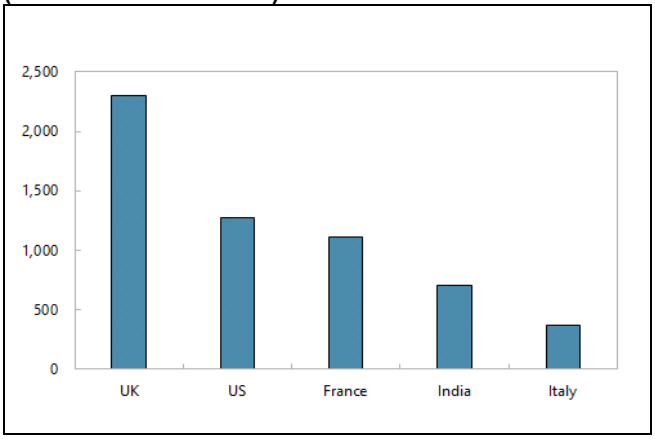
The presence of international banks in Kenya takes the form of foreign subsidiaries, foreign branches, or representative offices. Out of the 39 commercial banks in Kenya, 17 of them are foreign banks. As of December 2023, foreign banks in total own 30 percent of banking sector assets and 29.7 percent of deposits. Among the foreign banks, 14 institutions have the legal status of subsidiaries while the other 3 institutions are branches. The choice of foreign subsidiaries as the preferred legal status by most foreign banks reflects foreign banks' intention to establish large retail operations in Kenya. In addition, 9 other banks have opened representative offices in Kenya, which serve as marketing and liaison points for their parent banks and international affiliates.

Figure 2. International Bank Claims on Kenya
(Millions of US Dollars)



Source: Bank for International Settlements consolidated banking statistics

Figure 3. International Bank Claims on Kenya: Top Source Countries
(Millions of US Dollars)



Source: BIS consolidated banking statistics.
Note: The statistics include all sectors in Kenya and the split is based on nationality of the parent banking group. The data are for 2024 Q3 or the latest available period.

International banks' cross-border claims on Kenya are sizeable and broad-based. According to the CBK, foreign banks accounted for about 30 percent of total banking sector assets in Kenya at the end of 2024. Based on the BIS consolidated banking statistics[2], which capture the worldwide consolidated positions of internationally active banking groups headquartered in reporting countries, total cross-border bank claims on Kenya have declined moderately since 2021 and stood at around US$6.5 billion in 2024 (Figure 2)[3]. Such claims include debt instruments, equities, or investment fund shares. Banks from the UK, the US, France, India,

[2] The BIS consolidated banking statistics receive data from 31 major reporting economies and not all countries report data to the BIS. Therefore, the BIS numbers do not provide a full picture of cross-border banking linkages and should be interpreted with caution and considered jointly with the Central Bank of Kenya's banking supervision statistics describe above.

[3] In comparison, Kenya's total liabilities under the international investment position were about US$84 billion, which includes external liabilities vis-à-vis all foreign institutional sectors (general government, financial corporations, nonfinancial corporations, households, etc.).

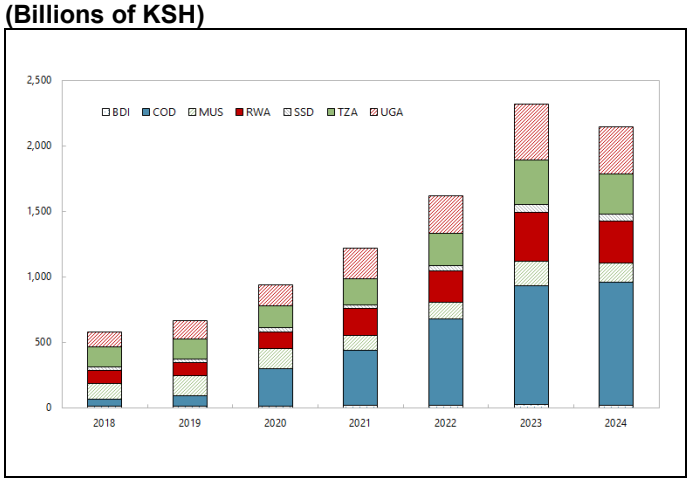
and Italy ranked among the top holders of financial claims on Kenya, reflecting the deep economic and cultural ties between Kenya and these nations (Figure 3).

2.2 Kenyan Banks' Outward Cross-Border Exposures to the Rest of Africa

In the meantime, Kenyan banks have a significant footprint in Africa, particularly within the East African Community. Kenyan banks' outward cross-border exposures take three forms. Destination countries where Kenyan banks operate through foreign subsidiaries or branches include Burundi, Congo DR, Mauritius, Rwanda, South Sudan, Tanzania, and Uganda. In Côte d'Ivoire and Ghana, Kenyan banks provide online services only. Lastly, Kenyan banks have taken ownership of local banks in Botswana, Malawi, Mozambique, and Zambia.

Kenyan banks' assets in other EAC countries have risen considerably in recent years. By the end of 2024, total assets by Kenyan banks in other EAC markets were KSH 2,147 billion, nearly 3.7 times the total assets held by Kenyan banks at end-2018 (Figure 4). Congo DR, which accounted for about 43% of Kenyan banks' total assets abroad in the EAC, is the largest destination market for Kenyan banks. Kenyan banks' linkages with Congo DR saw a remarkable shift, as Congo DR only represented a 10% share of Kenyan banks' total assets abroad within the EAC in 2018. Other notable destination markets for Kenyan banks are Rwanda, Tanzania, and Uganda, each accounting for around 15% of the total assets in 2024.

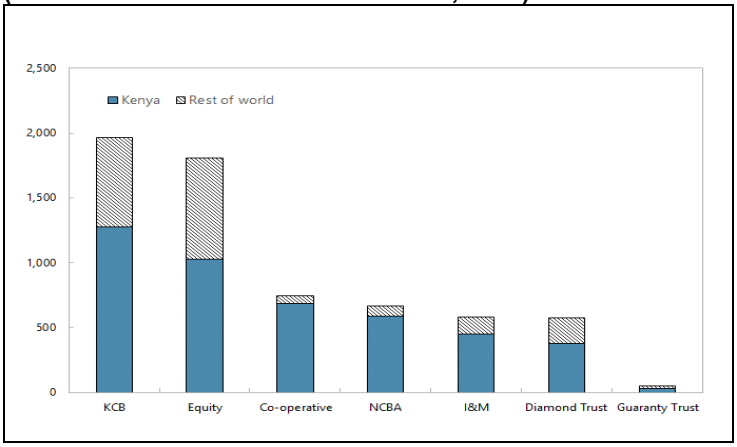
Figure 4. Kenyan Banks in the Region: Assets
(Billions of KSH)



Source: Central Bank of Kenya

KCB Group and Equity Bank Group are the two leading Kenyan banks in their overseas expansions. The list of Kenyan banks with regional expansions includes KCB, Equity Bank, Co-operative Bank, NCBA, I&M, Diamond Trust Bank, and Guaranty Trust Bank (Figure 5). As the two largest domestic banks in Kenya in terms of asset size, KCB and Equity Bank have taken the lion's share of the overseas assets in nominal terms, with KSH 685 billion and KSH 777 billion assets placed abroad, respectively. Other Kenyan banks such as Diamond Trust Bank, I&M, and NCBA have also placed considerable shares of their assets in foreign markets.

Figure 5. Asset Allocation of Kenyan Banks
(In billions of KSH as of December 31, 2024)

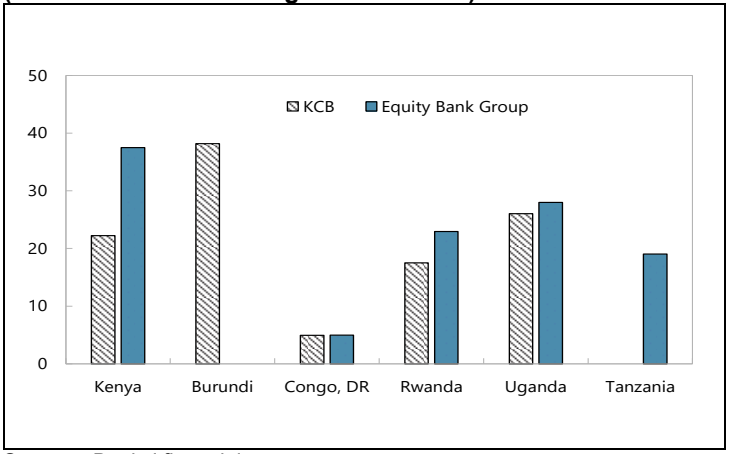


Sources: Central Bank of Kenya, Bank Supervision Report, 2024.

Kenyan banks have different sovereign debt securities exposures. Information based on KCB and Equity Bank's financial statements indicates that their securities holdings, which mainly comprise government debt securities, vary substantially across countries (Figure 6). The distinct levels of sovereign-bank nexus through sovereign debt holdings are probably because of relative attractiveness between sovereign yields and returns on lending activities, different degrees of perceived sovereign credit risks, and uneven banking sector regulatory requirements in destination countries.

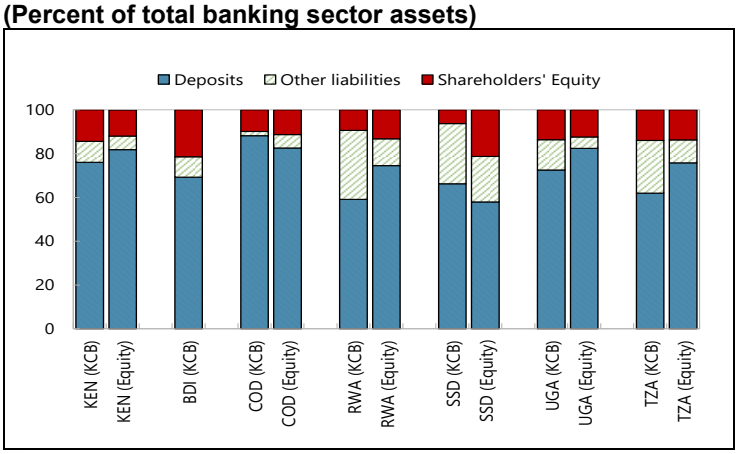
A noticeable feature of Kenyan banks' cross-border operations is their reliance on local funding sources. Overall, the two largest Kenyan banks have relied on a deposit-centric funding model across jurisdictions (Figure 7). Customer deposits are the predominant funding source, although there is some heterogeneity by bank and across countries. For instance, KCB depends less on deposits than Equity Bank to fund its business activities in Rwanda, South Sudan, Uganda, and Tanzania.

Figure 6. Kenyan Banks' Securities Holdings
(Percent of total banking sector assets)



Sources: Banks' financial statements.

Figure 7. Kenyan Banks' Funding Sources
(Percent of total banking sector assets)



Sources: Banks' financial statements.

2.3 Cross-Border Supervisory Efforts and Regional Banking Harmonization

The legal status of Kenyan banks in other EAC countries is universally foreign subsidiaries. Only the I&M Group has a joint venture in Mauritius, a nation outside the EAC. The legal form of cross-border banking operations has important implications for risk transmission and supervision. A subsidiary is a company owned or controlled by the parent company, operating as a separate legal entity for managing specific operations or limiting risks. Kenyan banks' choices of organizational form may reflect home-country regulations on how banks are allowed to operate overseas, regulatory restrictions and taxes in host countries, banks' intended degrees of penetration in the host market, parent banks' legal status of operations elsewhere, and risks that banks face in the host markets (Cerutti et al. 2007).

The Central Bank of Kenya has signed Memoranda of Understanding (MOUs) with multiple counterparts. Several MOUs were signed between the CBK and other central banks to provide formal arrangements regarding information exchange, joint inspections, and supervisory college attendance. According to the CBK banking supervision annual report, these are central banks from Burundi, Rwanda, Uganda, Tanzania, Mauritius, South Sudan, Malawi, Zimbabwe, Zambia, South Africa, Nigeria, Morocco, and India. While the existing MOUs cover most countries in which Kenyan banks have operations, agreements with home supervisors of international banks that have claims on Kenya are more limited. In total, the CBK has agreements with 4 home supervisors of international banks that have claims on Kenya and the CBK has not signed MOUs with any developed country counterparts despite substantial claims on Kenya by international banks from Western countries.

The CBK has also set up supervisory colleges for certain banks, but joint inspections have been infrequent. The CBK has established supervisory colleges for several major Kenyan banks. Nevertheless, the dates of last joint inspections with foreign supervisors vary across bank-country pairs, with most entries recording no joint inspection in the last ten years. Only one joint inspection is scheduled for KCB's foreign subsidiary in Rwanda in 2026.

Regional banking integration in the EAC is ongoing in terms of standards and crisis management frameworks. In EAC countries and Mauritius, where Kenyan banks have a notable footprint, heterogeneity can be found in accounting standards, capital definitions, and deposit insurance arrangements (Table 1). Kenya, Burundi, Mauritius, Rwanda, Tanzania, and Uganda have implemented or are moving towards International Financial Reporting Standards (IFRS) 9, whereas Congo DR and South Sudan use their national standards. Bank capital definitions also differ, with some jurisdictions aligned with Basel II/III and others not aligned with Basel II. Furthermore, the degrees of crisis management and financial safety net also vary in host markets where Kenyan banks operate, as not all countries have fully functional deposit insurance schemes. Uneven deposit insurance implementation and financial safety net coverage can increase uncertainty about burden sharing and depositor protection in a cross-border banking crisis (Drummond et al. 2015 and IMF 2015).

Table 1. Accounting Standards across Countries

Jurisdiction	Accounting standard	Capital definition	Deposit insurance
Kenya	IFRS 9	Basel I/II/III	Yes
Burundi	IFRS 9 in progress	Basel I/II	No
Congo DR	National	Not Basel II	No
Mauritius	IFRS 9	Basel II	Yes
Rwanda	IFRS 9	Basel II/III	Yes
South Sudan	National	Not Basel II	No
Tanzania	IFRS 9	Basel II/III	Yes
Uganda	IFRS 9	Basel II/III	Yes

Sources: IMF Financial Sector Stability Review (FSSR) and Financial Sector Assessment Program (FSAP) reports and authorities' websites.

Data and information gaps compound the challenges for cross-border banking supervision. The CBK has faced obstacles in compiling historical cross-border supervisory data at the bank level. In addition, existing databases from the International Monetary Fund and the World Bank reveal critical data gaps related to financial sector surveillance. For instance, there is no published data are available on capital adequacy or NPL ratios for South Sudan. Banking sector profitability indicators and loan-deposit spread data are also far from complete for Kenyan banks' main destination countries.

3. Empirical Analysis of Kenyan Banks' Overseas Expansions and Macroeconomic Spillover Effects

Kenya's large and rising cross-border exposures carry macroeconomic spillover effects. In this section, we address two research questions: What factors drive Kenyan banks' overseas asset growth? What are the macroeconomic effects of Kenyan banks' credit expansions in recipient countries?

3.1 Drivers of Kenyan Banks' Overseas Asset Growth

We rely on the Central Bank of Kenya's proprietary banking supervisory data[4] to conduct the analysis. The bank-level data from the CBK cover activities of 8 Kenyan banks (ABC Bank, Cooperative Bank, Diamond Trust Bank, Equity Bank, Guaranty Trust Bank, I&M Bank, KCB, and NCBA) in 7 countries (Burundi, Congo DR, Mauritius, Rwanda, South Sudan, Tanzania, and Uganda) over the period of 2017 Q1 to 2024 Q4.

We run panel data regressions with fixed effects to examine determinants of banks' overseas asset growth. As shown in Equation (1), the dependent variable is the year-over-year growth of bank i's assets measured in Kenyan shillings in host country j in time t. Following the banking literature (Cerutti et al. 2023; Correa et al. 2022; and Mathieu et al. 2019), explanatory variables include macroeconomic factors (real GDP growth, CPI inflation, bilateral trade flows, and public debt to GDP ratios) and banking sector-specific factors (return on assets, return on equity, and NPL ratios). The choice of control variables is subject to data constraints for sample countries. The regression setup also contains bank, country, and time fixed effects. Details on variable description and data sources are provided in the annex.

$$y_{i,j,t} = BX + \propto_i + \gamma_j + \mu_t + \epsilon_{i,j,t} \quad (1)$$

Results suggest that Kenyan banks' overseas asset growth is positively associated with bilateral trade flows and destination countries' banking sector fundamentals (Table 2). On average, a one percentage point increase in Kenya's lagged export growth to destination countries is associated with a 0.2 percentage point rise in Kenyan banks' asset growth abroad. In addition, destination markets' banking sector profitability and asset quality matter. Kenyan banks are more likely to expand assets in markets where returns on assets are high or NPL ratios are low. Other macroeconomic factors such as real GDP growth, CPI inflation, and public debt to GDP ratios do not have statistically significant explanatory power on Kenyan banks' overseas asset growth.

Table 2. Determinants of Kenyan Banks' Cross-Border Asset Expansion

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Kenyan banks' overseas assets, y/y growth					
Real GDP growth in destination country	-0.689 (0.494)	-0.695 (0.576)	-0.860 (0.520)	-0.793 (0.585)	-0.694 (0.620)
CPI inflation in destination country	0.116 (0.248)	0.217 (0.267)	0.533 (0.430)	0.603 (0.480)	1.829 (1.134)
Kenya's exports to destination country, lagged	0.202** (0.0836)	0.197** (0.0804)	0.169* (0.0911)	0.170* (0.0872)	0.229** (0.106)
Public debt to GDP ratio in destination country, lagged		-0.211 (0.318)			
Banking sector ROA in destination country, lagged			4.280* (2.385)		
Banking sector ROE in destination country, lagged				0.526 (0.416)	

[4] Only this empirical analysis section uses the CBK's proprietary banking supervisory data. The rest of the sections in this paper utilize information from the CBK's bank supervision annual reports, which are published once a year online.

Banking sector NPL ratio in destination country, lagged					-3.209* (1.636)
Constant	18.51*** (3.644)	29.04* (16.83)	6.063 (10.66)	9.096 (9.691)	30.89*** (9.267)
Number of observations	590	541	446	446	418
R-squared	0.129	0.117	0.138	0.119	0.172
Robust standard errors in parentheses. ***p<0.01, **p<0.05, * p<0.1.					

3.2 Macroeconomic Effects of Kenyan Banks' Credit Expansions in Recipient Countries

We investigate the implications of bank credit expansion on economic growth and inflation. Kenya had strong real GDP growth in recent years, averaging 5.6 percent during 2021-2025 and above the Sub-Saharan Africa's average of 4.2 percent. During the same period, CPI inflation was 6 percent on average and consistently stayed within the target band since the adoption of inflation targeting in 2020.[5] Meanwhile, increased external financial linkages generate spillover effects for developing countries (Cubeddu et al. 2014) and managing external volatilities can be challenging for non-reserve issuing economies (Poirson et al. 2022). A large shock to banks' external funding can negatively affect banks' domestic lending (Aiyar 2012), which is a critical channel for transmitting funding conditions of parent banks in Kenya to credit supplies in other markets where Kenyan banks operate.

We use a dynamic panel data model to study the contemporaneous effects of bank lending. To overcome data limitations, we choose a broad country sample of 32 African countries using annual data between 2003 and 2024.[6] As economic growth and inflation tend to have inertia, we follow the economic literature and apply the following dynamic panel data model in Equation (2):

$$y_{i,t} = \varphi y_{i,t-1} + \beta_1 TBC_{i,t} + \beta_2 KBC_{i,t} + \gamma X + \alpha_i + \mu_t + \epsilon_{i,t} \quad (2)$$

The dependent variable y is either real GDP growth or CPI inflation in country i at time t. Growth of total bank credit (TBC) and Kenyan bank credit (KBC) to the private sector are the key explanatory variables of our interest.[7] The total bank credit and Kenyan bank credit impulses are defined as changes in credit to the private sector in percent of previous year's nominal GDP by all banks and Kenyan banks, respectively. The set of control variables follows the economic literature on determinants of economic growth and inflation (Acemoglu et al. 2003; Barro 2003; Berg et al. 2018; Zhang and Wang 2022). Specifically, control variables in real GDP regressions are population growth, human capital growth, and institutional quality. Control variables in CPI inflation regressions include nominal effective exchange rate and terms of trade.[8] To address bias in dynamic panel data estimations, the Generalized Method of Moments (GMM) is applied. The lagged dependent variable

[5] For a detailed discussion on recent macroeconomic developments in Sub-Saharan Africa and comparison between actual inflation and inflation targets and bands, please refer to IMF (2026) and Zhang (2025).

[6] Choice of annual data and country coverage is based on data quality and availability. Quarterly data are not used as half of the African countries do not publish quarterly GDP series. We also exclude several countries from our sample due to concerns of their national accounts statistics. The sample country list is included in the annex.

[7] B_2 is the differential effect of replacing one unit of other bank credit with one unit of Kenyan bank credit in a host country, holding the total amount of credit unchanged. The total marginal effect of Kenyan bank credit on macroeconomic variables is $B_1 + B_2$.

[8] Given the sample has 32 African countries, our choice of control variables is constrained by data availability.

is the GMM-style instrument and time dummy variables are used as IV-style instruments. Country and time fixed effects are also included.

We deploy the local projection method of Jorda (2005) to examine future effects of bank credit growth. The dependent variable for this part of the analysis is the h-year ahead either real GDP growth or CPI inflation so that we can examine responses of macroeconomic performance in h periods ahead to bank credit growth at time t. The control variables remain the same, with country and time fixed effects included in Equation (3):

$$y_{i,t+h} = \varphi^h y_{i,t-1} + \theta^h BC_{i,t} + \beta_1^h TBC_{i,t} + \beta_2^h KBC_{i,t} + \gamma^h X + \alpha_i^h + \mu_i^h \mu_t + \epsilon_{i,t} \quad (3)$$

Table 3. Effect of Bank Credit Growth on Contemporaneous Real GDP Growth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable: real GDP growth								
Lagged real GDP growth	0.161** (0.0661)	0.154* (0.0850)	0.225** (0.0912)	0.226** (0.0920)	0.103 (0.0628)	0.176** (0.0842)	0.243*** (0.0876)	0.230** (0.0932)
Total bank credit growth	-0.0181 (0.0108)	0.238*** (0.0853)	0.222** (0.0946)	0.223** (0.0949)	0.0381 (0.0826)	1.145** (0.557)	1.077* (0.578)	1.086* (0.600)
Kenyan bank credit growth				-0.357 (0.244)				0.486 (3.981)
Population growth	1.409** (0.588)	1.635** (0.696)	1.437* (0.801)	1.440* (0.800)	0.500 (0.698)	1.166 (1.020)	2.276 (1.454)	2.027* (1.194)
Human capital growth		0.326 (0.352)				1.404 (0.988)		
Institutional quality change			0.331*** (0.0634)	0.332*** (0.0634)			0.570*** (0.131)	0.589*** (0.135)
Constant	-1.792 (1.089)	22.57** (9.015)	21.50** (9.553)	21.51** (9.574)	6.409 (9.148)	113.1** (56.00)	104.4* (58.44)	106.0* (60.05)
Number of observations	642	524	470	470	642	524	470	470
Number of countries	32	27	24	24	32	27	24	24
Estimation method	Panel FE	Panel FE	Panel FE	Panel FE	GMM	GMM	GMM	GMM
R-squared	0.063	0.075	0.164	0.164	N.A.	N.A.	N.A.	N.A.
Arellano-Bond AR(1) test, p value	N.A.	N.A.	N.A.	N.A.	0.001	0.001	0.002	0.002
Arellano-Bond AR(2) test, p value	N.A.	N.A.	N.A.	N.A.	0.136	0.954	0.221	0.225
Hansen identifying restrictions test, p value	N.A.	N.A.	N.A.	N.A.	1.000	1.000	1.000	1.000
Robust standard errors in parentheses. *** p<0.01, ** p<0.05. * p<0.1.								

Table 4. Effect of Bank Credit Growth on Contemporaneous CPI Inflation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable: CPI inflation								
Lagged CPI inflation	0.218*** (0.0525)	0.221*** (0.0539)	0.219*** (0.0530)	0.213*** (0.0528)	0.436*** (0.0569)	0.422*** (0.0536)	0.410*** (0.0607)	0.391*** (0.0571)
Total bank credit growth	0.0478** (0.0197)	0.0339* (0.0184)	0.0328* (0.0179)	0.0281 (0.0182)	0.183*** (0.0520)	0.176*** (0.0512)	0.149*** (0.0516)	0.156*** (0.0512)
Kenyan bank credit growth				2.310*** (0.818)				4.197** (1.632)
Nominal effective exchange rate, percent change		-0.127*** (0.0398)	-0.129*** (0.0400)	-0.132*** (0.0395)		-0.0857* (0.0474)	-0.0978** (0.0498)	-0.0779 (0.0591)
Terms of trade growth			0.0114 (0.0155)	0.0113 (0.0157)			0.0726 (0.0486)	0.0716 (0.0665)
Constant	9.722*** (1.652)	8.110*** (1.503)	7.982*** (1.454)	7.355*** (1.538)	21.74*** (5.147)	20.94*** (5.079)	18.08*** (5.155)	18.62*** (5.056)
Number of observations	504	504	504	504	504	504	504	504
Number of countries	32	32	32	32	32	32	32	32
Estimation method	Panel FE	Panel FE	Panel FE	Panel FE	GMM	GMM	GMM	GMM
R-squared	0.061	0.114	0.115	0.143	N.A.	N.A.	N.A.	N.A.
Arellano-Bond AR(1) test, p value	N.A.	N.A.	N.A.	N.A.	0.001	0.001	0.001	0.001
Arellano-Bond AR(2) test, p value	N.A.	N.A.	N.A.	N.A.	0.356	0.339	0.141	0.361
Hansen identifying restrictions test, p value	N.A.	N.A.	N.A.	N.A.	0.999	0.999	1.000	0.999
Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.								

Empirical results point to positive effects of bank lending on real GDP growth and inflation (Tables 3 and 4).[9] As expected, the lagged dependent variables are positive and statistically significant and the GMM diagnostic statistics look reasonable, validating the use of this dynamic model. Across this broad sample of African economies, a one percentage point change in total bank credit to the private sector is associated with a 1.1 percentage point increase in real GDP growth. In addition, population growth and improvements in institutional quality are positively correlated with economic growth. Similarly, a one percentage point change in total bank credit is associated with a 0.2 percentage point increase in CPI inflation. Moreover, nominal effective exchange rate appreciation is strongly linked to lower domestic inflation. Kenyan banks' credit to the private

[9] Given the potential endogeneity of credit supply and demand and the limited availability of comparable banking data across African economies, the estimates should be interpreted as conditional associations rather than causal effects.

sector is found not statistically different than other banks' credit in the relation to real GDP growth in host countries. However, Kenyan banks' credit to the private sector is more positively associated with CPI inflation in host countries, given the estimated coefficient is positive and the statistical different from zero. The estimated coefficients of the interaction terms between bank credit growth and Kenyan banking destination are statistically insignificant, indicating that there is no statistical difference in the macroeconomic effects of bank credit growth between markets where Kenyan banks have a presence and other markets.

Results of impulse responses suggest that bank lending growth does not affect future output but not inflation (Figures 8 and 9). The response of real GDP growth to an increase in total bank lending growth is not statistically different from zero as the confidence intervals stay within the horizontal zero line for throughout the projection horizon. On the other hand, the response of CPI inflation to total bank lending growth does not statistically differ from zero as the confidence intervals include the horizontal zero line the next two years. Therefore, we find that bank lending growth has short-lived effects and are only positively associated with higher real GDP growth and CPI inflation in the current year and do not persist in the next few years.

Figure 8. Effect of Bank Credit Growth on Future Real GDP Growth

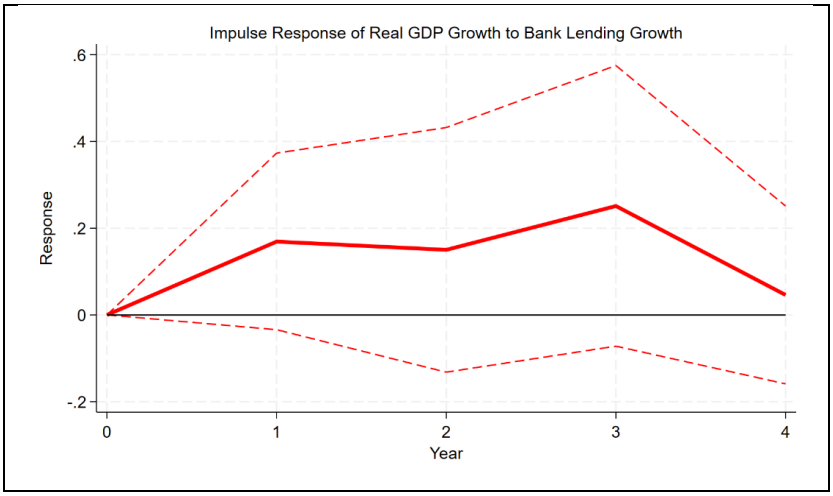
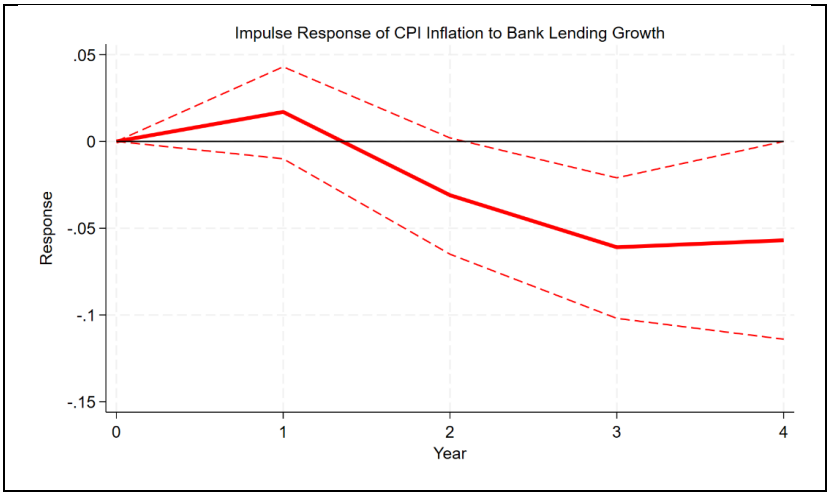


Figure 9. Effect of Bank Credit Growth on Future CPI Inflation



4. Robustness Checks

We perform several robustness checks of our empirical results (Tables 5 and 6). First, we exclude episodes of systemic banking crises based on the database by Laeven and Valencia (2013), who identify dates, costs, and policy interventions associated with banking crises on a cross-country basis since 1970. The results on real GDP growth and CPI inflation remain broadly consistent with the baseline regression results, and our findings do not appear to be driven by historical banking crises in host countries. Second, we split our sample into bank credit expansion periods to analyze whether macroeconomic variables respond asymmetrically to bank credit growth. The estimated coefficients on total bank credit growth in real GDP regressions are still statistically significant and positive during bank credit expansions. The effects of total bank credit growth on CPI inflation, on the other hand, are statistically insignificant during bank credit expansions. Furthermore, we drop periods in which real GDP growth is negative. In line with our baseline results, the effects of total bank lending growth are estimated to be statistically positive on both output and inflation, eliminating the possibility that results are affected by a few years of negative economic growth. The last check is to exclude deflationary periods from the sample. Results suggest that bank lending growth is positively associated with CPI inflation but not real GDP growth during inflationary periods. Overall, these robustness checks indicate that the positive effects of bank lending growth on real GDP and CPI inflation are broadly consistent and statistically significant across different economic regimes and conditions.

Table 5. Robustness Check: Effect of Bank Credit Growth on Real GDP Growth

	(1)	(2)	(3)	(4)
Dependent variable: real GDP growth				
Lagged real GDP growth	0.171**	0.147*	0.0962	0.229*
	(0.0849)	(0.0777)	(0.0925)	(0.119)
Total bank credit growth	1.075*	1.497***	1.259**	1.023
	(0.556)	(0.580)	(0.622)	(0.708)
Kenyan bank credit growth	0.407	1.923	1.925	0.628
	(3.482)	(3.867)	(3.924)	(3.458)
Population growth	2.428*	1.041	2.096*	1.894
	(1.310)	(1.093)	(1.104)	(2.691)
Institutional quality change	0.559***	0.439**	0.390**	0.547***
	(0.145)	(0.206)	(0.181)	(0.146)
Constant	104.1*	150.4***	123.8**	100.0
	(56.15)	(57.17)	(62.69)	(68.74)
Sample	Non banking crisis	Bank credit expansion	Positive GDP growth	Positive inflation
Number of observations	452	287	422	446
Number of countries	24	24	24	24
Estimation method	GMM	GMM	GMM	GMM
Arellano-Bond AR(1) test, p value	0.002	0.014	0.005	0.004
Arellano-Bond AR(2) test, p value	0.322	0.087	0.396	0.238
Hansen identifying restrictions test, p value	1.000	1.000	1.000	1.000
Robust standard errors in parentheses. *** p<0.01, ** p<0.05. * p<0.1.				

Table 6. Robustness Check: Effect of Bank Credit Growth on CPI Inflation

	(1)	(2)	(3)	(4)
Dependent variable: CPI inflation				
Lagged CPI inflation	0.388*** (0.0661)	0.344*** (0.132)	0.384*** (0.0531)	0.379*** (0.0600)
Total bank credit growth	0.157*** (0.0570)	-0.243 (0.413)	0.162*** (0.0538)	0.218** (0.105)
Kenyan bank credit growth	4.517*** (1.588)	2.547** (1.065)	4.613*** (1.673)	4.017** (1.745)
Nominal effective exchange rate, percent change	-0.0931* (0.0478)	-0.0594 (0.0533)	-0.0690 (0.0431)	-0.0979 (0.0630)
Terms of trade growth	0.0392 (0.0503)	0.0869 (0.0594)	-0.0146 (0.0738)	0.0615 (0.0671)
Constant	18.72*** (5.631)	-21.26 (40.84)	19.42*** (5.404)	25.42** (10.50)
Sample	Non banking crisis	Bank credit expansion	Positive GDP growth	Positive inflation
Number of observations	481	330	462	457
Number of countries	32	32	32	32
Estimation method	GMM	GMM	GMM	GMM
Arellano-Bond AR(1) test, p value	0.002	0.016	0.001	0.005
Arellano-Bond AR(2) test, p value	0.710	0.604	0.686	0.629
Hansen identifying restrictions test, p value	0.999	1.000	0.999	1.000
Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.				

5. Discussion and Conclusion

This paper documents the scale and composition of Kenya's cross-border banking exposures and highlights their systemic importance. Kenya is in a special position: it hosts several large international banks and also witnesses its home-grown banks venture into other African countries. Kenya's cross-border banking diversification to other parts of Africa is significant and has increased over time, with dominant market shares in several countries within the EAC. Their wide-ranging geographic presence and remarkable cross-border exposures are of systemic importance to Africa because of interconnectedness among African banking systems and heightened sovereign-bank linkages.

Cross-border banking activities create opportunities and risks, and we provide evidence on the factors associated with Kenyan banks' overseas expansion and the macroeconomic importance of the bank-credit channel. While supporting economic growth, employment, and financial inclusion during normal times, negative shocks in cross-border bank lending tighten credit conditions for borrowers and constrain investment (Hale et al. 2020). Kenyan banks' overseas expansions go hand in hand with Kenya's bilateral exports and host countries' banking sector profitability and asset quality. Kenyan banks' outward activities can generate macroeconomic spillovers through the bank lending channel, as our findings indicate slower bank credit growth is closely tied to lower local economic output and inflation. Furthermore, a financial crisis triggered by a

systemic bank failure could have even greater impacts on the economy beyond the bank lending channel. According to Laeven and Valencia (2013), the median fiscal cost for systemic banking crises is 10 percent of GDP for low and middle-income countries. However, one reassuring fact regarding financial stability is that Kenyan banks' activities in the EAC region are primarily funded by customer deposits in local markets and such regional banking integration partly reflect trade-based economic integration.

Our findings highlight the importance of cross-border banking linkages and spillover effects and raise additional questions for future research. Due to data constraints, we are unable to quantify and include the relevant variables of cross-border banking supervision and crisis management in the empirical analysis. For instance, do home-host coordination efforts such as expanded MOU coverage, regular joint inspections, strong supervisory college practices, and the establishment of a formal protocol for cross-border banking crisis resolution limit the frequency and severity of banking crises? Moreover, what is the role of regional banking harmonization—through the convergence of accounting standards, bank capital definitions, and deposit insurance arrangements toward international best practices—in shaping cross-border banking exposures and containing financial vulnerabilities? We leave these ideas for future research.

Annex 1. Additional Stylized Facts of Kenya's Cross-Border Banking Exposures

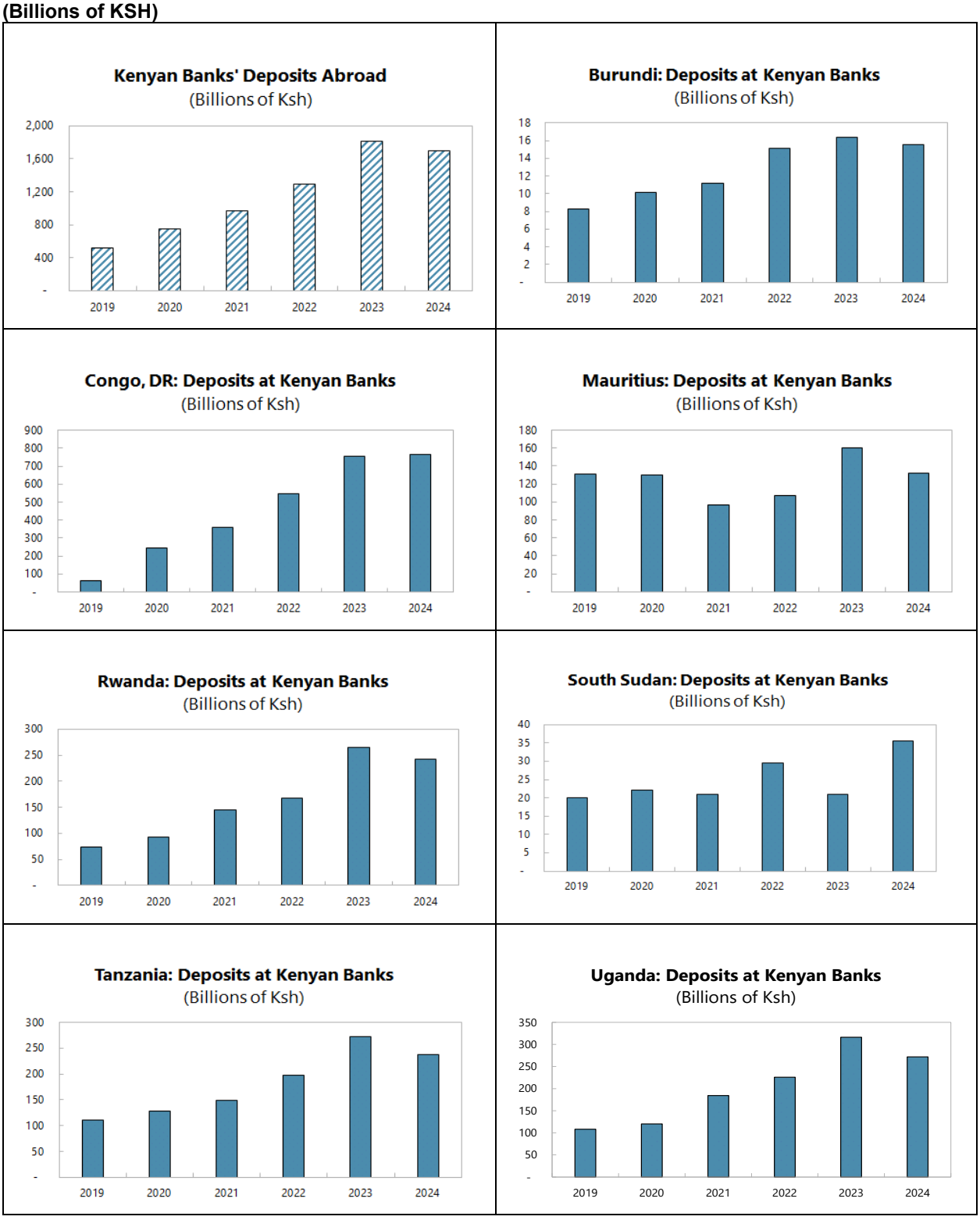
Kenyan banks' deposits have been on an upward trend in most destination countries (Appendix Figure 1). Total bank deposits by Kenyan banks abroad, the main component of liabilities, closely resemble the patterns of bank assets. In all destination markets except Mauritius, bank deposits have generally increased. The most striking ascent in bank deposits occurred in Congo DR, where customer deposits jumped more than tenfold.

The increases in the number of bank branches and employees in Kenyan banks' destination countries have been more modest (Appendix Figure 2). While Kenyan banks did open more branches and hire additional employees, their growth dynamics were not strong when compared with asset and deposit patterns. The relatively moderate additions in bank branches and employees could be a result of digitalization, as banks switch more services from offline to online and reduce the need to maintain heavy physical branches and staff.

Kenyan banks have dominated several African markets in terms of market shares (Appendix Figure 3). When measured in percent of total banking sector assets, Kenyan banks' market shares are not only the largest at home but also overwhelmingly sizeable in Congo DR, Rwanda, and South Sudan: KCB and Equity Bank are the second- and third-largest banks in these countries. In addition, several Kenyan banks are among the top 10 banks in Tanzania and Uganda. From the perspective of host countries, Kenyan banks are systemically important due to their large asset shares and deposit bases. In total, Kenyan banks hold nearly 65% of banking sector assets in Rwanda, 48% of banking sector assets in Congo DR, and 29% of banking sector assets in Uganda (Appendix Figure 4).

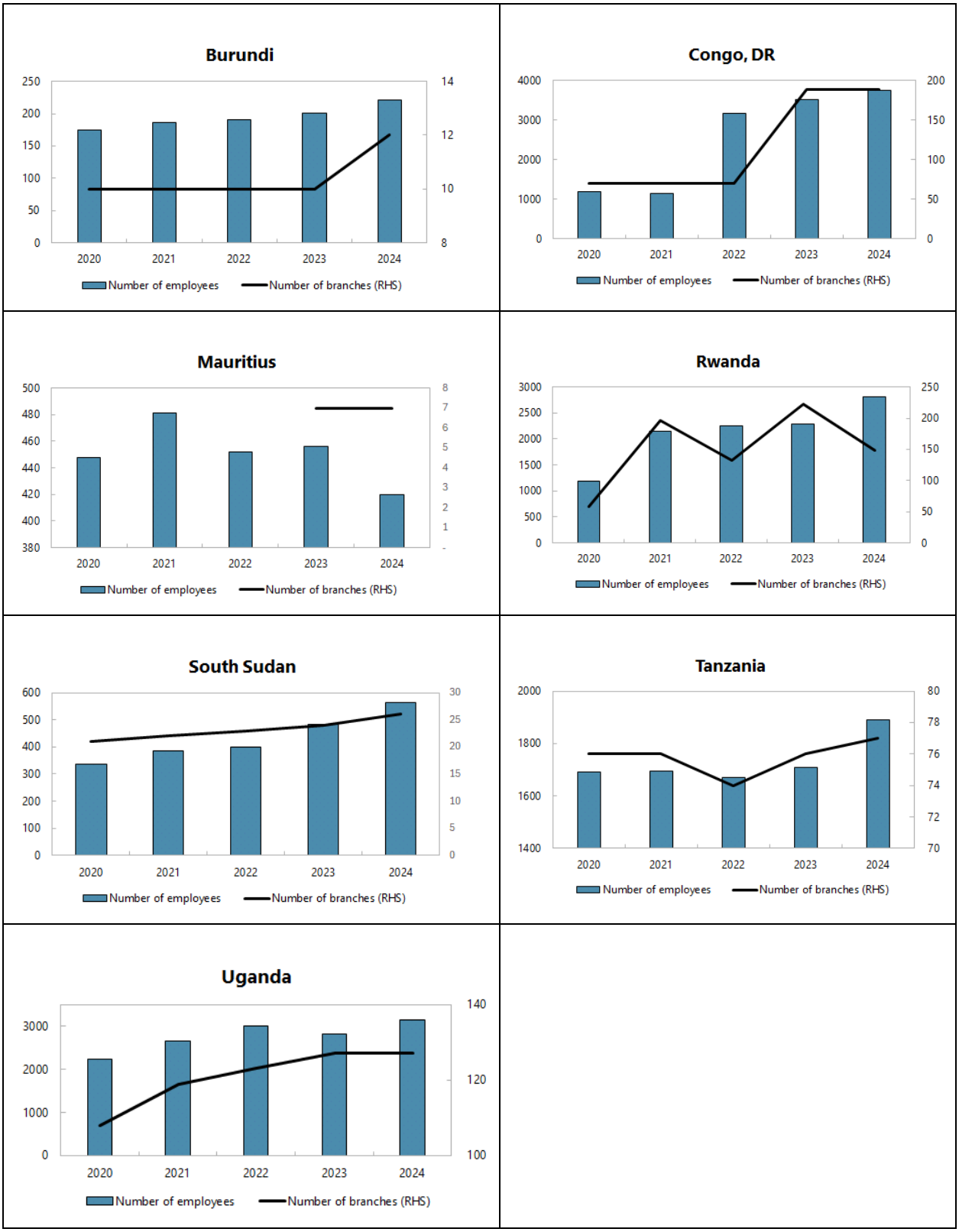
Kenyan banks' asset quality appears to be lower than industry averages in several destination countries. KCB and Equity Bank's country-specific non-performing loan (NPL) ratios are higher than sector averages in Kenya, Rwanda, and Uganda (Appendix Figure 5). This represents a pocket of risk given the combined significant market shares and suboptimal credit quality. Furthermore, elevated NPL ratios in certain locations as opposed to others within the same banking group could be a confluence of several factors, including regulatory and accounting standards adopted by local jurisdictions, enforcement of NPL regulations by bank supervisors, and strictness of banks' own loan quality recognition practices.

Appendix Figure 1. Kenyan Banks in the Region: Assets by Country (Billions of KSH)



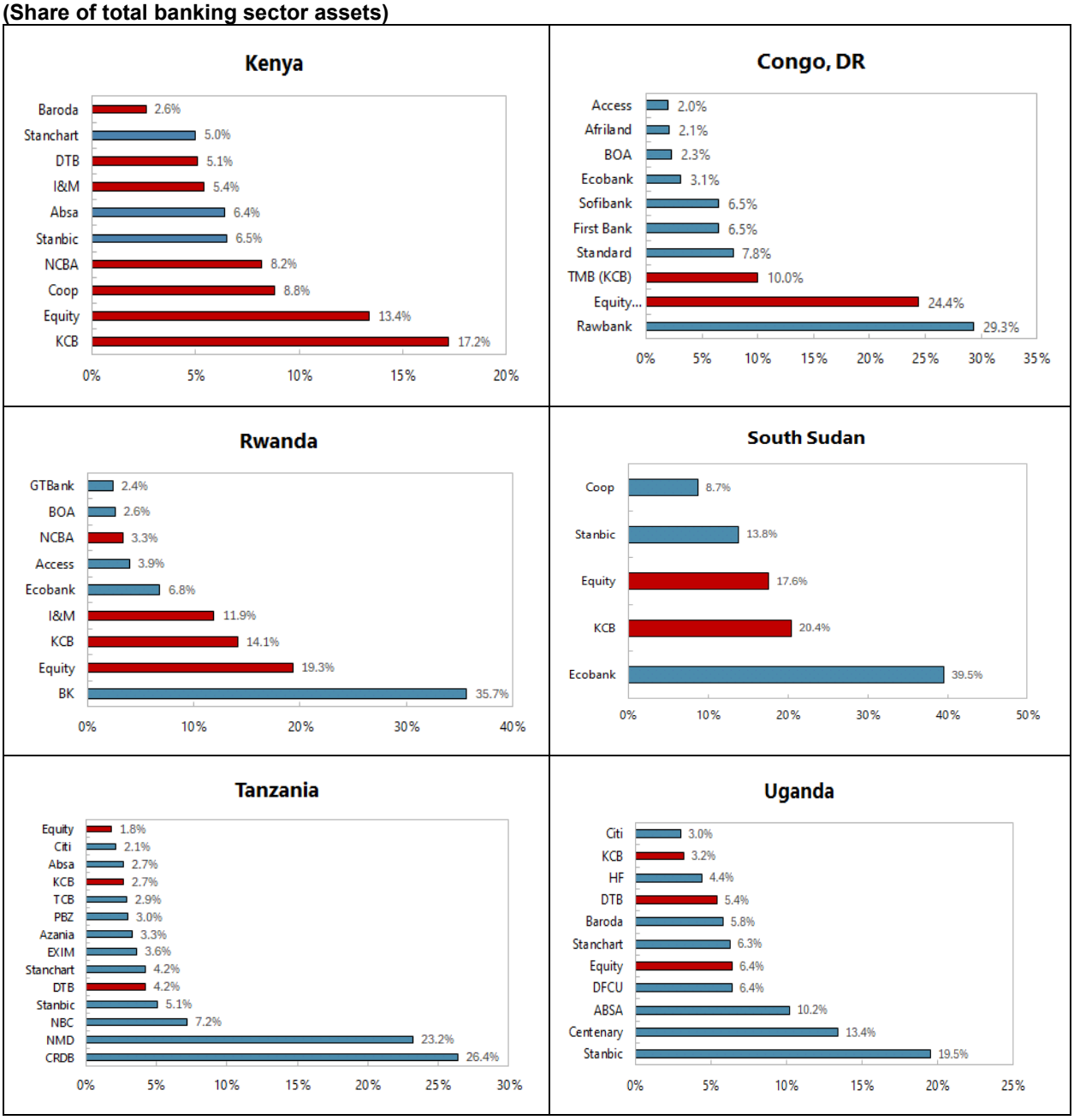
Sources: Central Bank of Kenya, Bank Supervision Annual Reports.

Appendix Figure 2. Bank Branches and Employees in Each Destination Country



Source: Central Bank of Kenya, Bank Supervision Annual Reports.

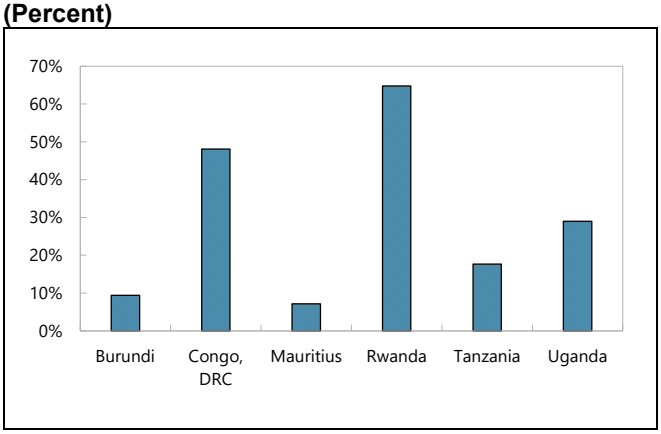
Appendix Figure 3. Kenyan Banks in the Region
(Share of total banking sector assets)



Sources: Equity Group Holdings, Investor Briefing FY 2025 performance, Central Banks, Bank Financial Statements.

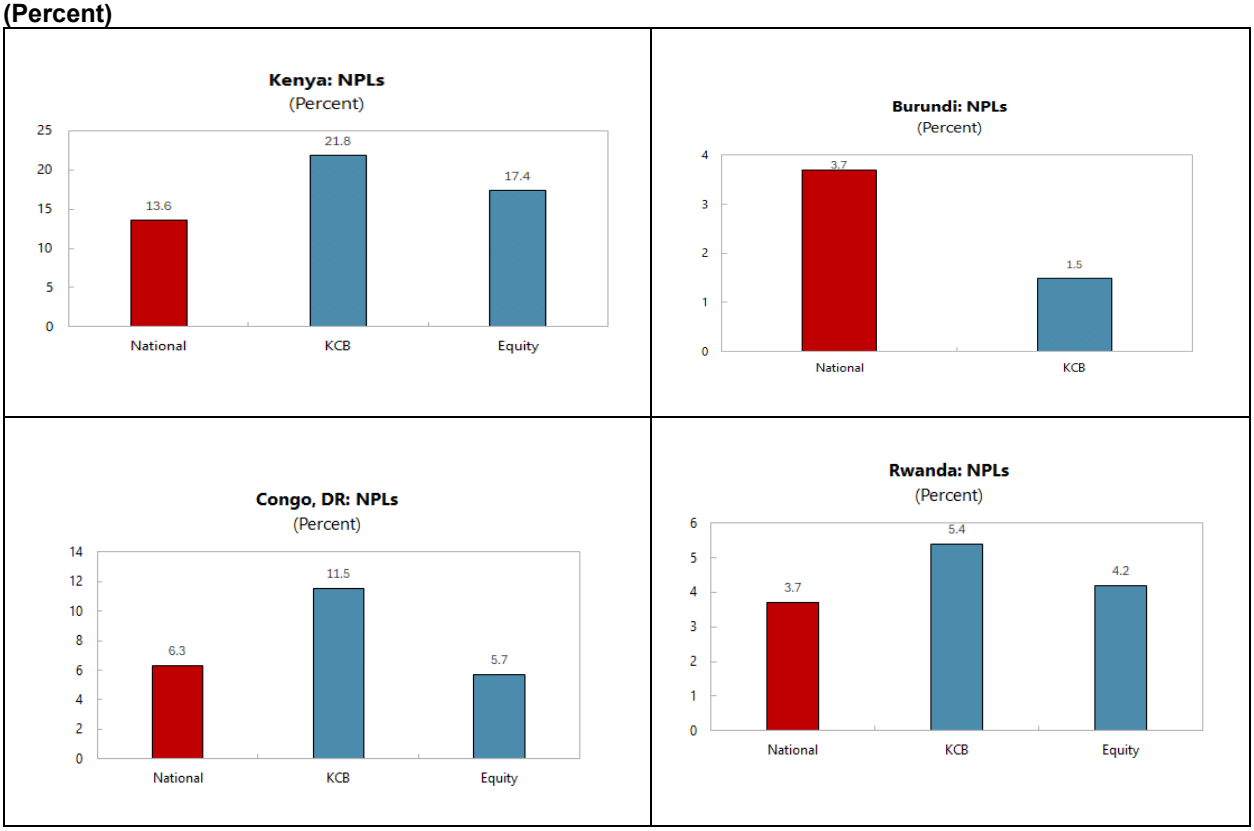
Note: Red columns denote Kenya-headquartered banks, and blue columns denote banks that have headquarters elsewhere.

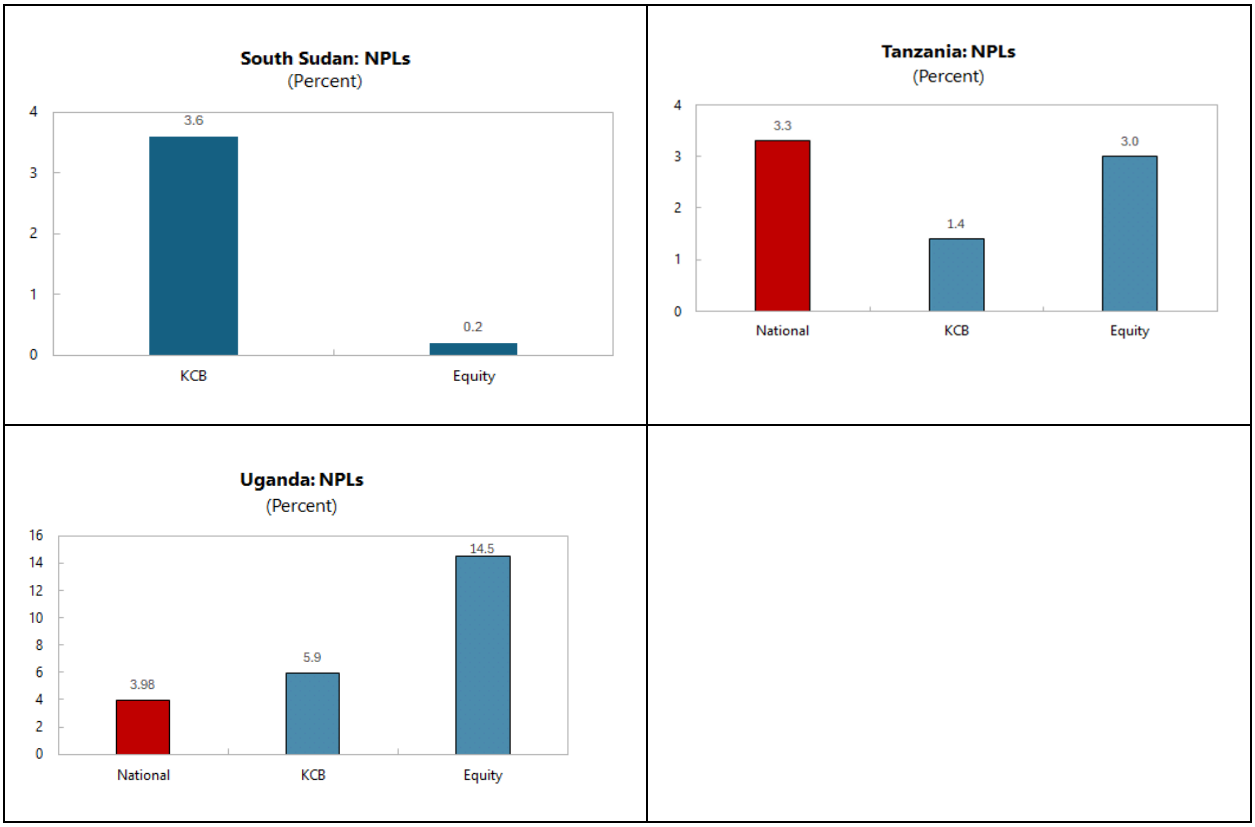
Appendix Figure 4. Kenyan Banks' Asset Shares in Destination Countries (Percent)



Sources: Central Bank of Kenya banking supervision statistics, World Economic Outlook, and IMF staff estimates.

Appendix Figure 5. Non-Performing Loans per Country (Percent)





Sources: Central Banks, KCB and Equity Annual Reports.

Annex 2. Kenyan Banks' Legal Status Overseas

Appendix Table 1. Legal Status by Kenyan Banks Across Destination Countries

	KCB Group	Equity Group	Diamond Trust Bank Group	NCBA Group	I&M Group	Guaranty Trust Bank	Cooperative Bank	ABC Bank
Burundi	Subsidiary		Subsidiary					
Congo DR	Subsidiary	Subsidiary						
Mauritius					Joint Venture			
Rwanda	Subsidiary	Subsidiary		Subsidiary	Subsidiary	Subsidiary		
South Sudan	Subsidiary	Subsidiary					Subsidiary	
Tanzania	Subsidiary	Subsidiary	Subsidiary	Subsidiary	Subsidiary			
Uganda	Subsidiary	Subsidiary	Subsidiary	Subsidiary	Subsidiary	Subsidiary		Subsidiary

Source: Central Bank of Kenya

Note: A subsidiary is a company owned or controlled by the parent company, operating as a separate legal entity for managing specific operations, entering new markets, or limiting risk.

Annex 3. Description of Data Used in Empirical Analysis

Sample Country List for the Analysis of Macroeconomic Effects of Bank Lending

Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Chad, Comoros, Congo Republic, Côte d'Ivoire, Ethiopia, Gabon, Gambia, Ghana, Guinea, Madagascar, Malawi, Mali, Mauritius, Mozambique, Namibia, Niger, Nigeria, Rwanda, Senegal, Seychelles, South Africa, Tanzania, Togo, Uganda, Zambia.

Variable Description and Data Sources

Variables Used to Analyze Determinants of Bank Lending

Variable	Description	Data source
Cross-border bank assets	Assets of a particular Kenyan bank in a given country	Central Bank of Kenya
Real GDP growth	GDP in constant prices, percent change	IMF World Economic Outlook
CPI inflation	Consumer price index, period average percent change	IMF World Economic Outlook
Kenya's bilateral exports	Kenya's exports vis-à-vis another country	United Nations Comtrade Database
Public debt to GDP ratio	General government gross debt in percent of GDP	IMF World Economic Outlook
Banking sector ROA	Banking sector return on assets	IMF Financial Soundness Indicators
Banking sector ROE	Banking sector return on equity	IMF Financial Soundness Indicators
Banking sector NPL	Banking sector non-performing loans to gross loans	IMF Financial Soundness Indicators

Variables Used to Analyze Macroeconomic Effects of Bank Lending

Variable	Description	Data source
Real GDP growth	GDP in constant prices, percent change	IMF World Economic Outlook
CPI inflation	Consumer price index, period average percent change	IMF World Economic Outlook
Bank credit growth	Change in bank claims on private sector, in percent of previous year nominal GDP	IMF Monetary and Financial Statistics
Population growth	Population, percent change	IMF World Economic Outlook
Human capital growth	Human capital index, percent change	Penn World Table
Terms of trade growth	Terms of trade index, percent change	IMF World Economic Outlook
Institutional quality change	Composite rating of economic, financial, and political risks, percent change	International Country Risk Guide
Nominal effective exchange rate growth	Nominal effective exchange rate index, percent change	IMF Effective Exchange Rate

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PUBLICATIONS

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